

Mak Chun Chi

</> Analyst Programmer · 6+ years experience

 +852 6340 0077

 martinmcc5804@gmail.com

 ISC2 CISSP Study Notes & Q&A
(Accreditation to *Deepseek* & *Manus*)

 CompTIA Network+ (N10-009) Study
Notes (Accreditation to *Deepseek* & *Manus*)

⚙️ CORE COMPETENCIES

Languages

C#

Java

JavaScript

HTML5/CSS3

PHP

Frameworks

.NET

Spring Boot

React

Node.js

Express

Databases

MySQL

SQL Server

DevOps & Tools

Docker

Git

REST APIs

Socket Programming

WordPress

Expertise

Root Cause Analysis

Technical Documentation

Input Validation

Legacy Modernization

🎓 EDUCATION

Bachelor of Engineering in Electrical Engineering
The Hong Kong Polytechnic University · 2014 – 2016

Higher Diploma in Electrical Engineering
The Hong Kong Polytechnic University · 2012 – 2014

⚙️ CERTIFICATIONS

🏆 [CompTIA Network+ \(N10-009\) – 2026](#)

⚙️ CERTIFICATE OF COMPLETION

🔒 [ISC2 CISSP \(Udemy\) – 2026](#)

📶 [CompTIA Networking+ \(Udemy\) – 2026](#)

🧠 ACADEMIC PROJECTS

Solar Energy Optimization System (Bachelor Thesis)
Optimised solar panel efficiency through load analysis and tracking algorithms (PolyU 2015-2016).

Embedded Line-follower Robot (Higher Diploma)
Developed a driverless car using sensor fusion and microcontroller logic (PolyU 2013-2014).

👤 **Analyst Programmer** with **6+ years of full-stack development** and system maintenance expertise. Specialised in **C# .NET, Java Spring Boot, LAMP, React/Node.js**. For the past four years, I have built and maintained **earthquake information systems for the Hong Kong Observatory**, driving Docker containerisation, technical documentation, and root cause analysis to enhance platform reliability.

💼 EMPLOYMENT & PROJECT EXPERIENCE

Analyst Programmer @ IT Solutions Ltd. (Hong Kong Observatory)

Jul 2025 – Present

- Developed a **React + Express + Node.js** web application to collect and visualise locally felt earthquake reports, deployed with **Docker** inside a secured internal network.
- Authored technical documentation and user guides; performed proactive root cause analysis and preventive fixes to reduce recurring incidents.

Analyst Programmer @ Kelly Services (Hong Kong Observatory)

Jun 2022 – Jun 2025

- Built a full-stack earthquake data dissemination platform using **PHP/LAMP** stack running inside Docker; enforced client-side validation.
- Led **root cause analysis (RCA)** initiatives and implemented preventive patches, significantly reducing recurrence of system failures.
- Managed WordPress-based technical documentation hub; standardised user guides for 3 internal applications, adopted as team-wide reference.

Solution Developer @ PCCW

Sep 2019 – Jun 2022

- Developed a web-based **booking appointment system** using **C# .NET**, supporting real-time availability validation, rescheduling, and cancellations.
- Built a **Windows Forms** desktop application as a query interface for backend databases.
- Implemented a **multithreaded socket server in Java** for concurrent client connections in real-time data exchange.
- Collaborated on a **Spring Boot workflow management system**.

🏆 KEY ACHIEVEMENTS



Containerisation Migration ⚡ **IMPACT: DEPLOYMENT CONSISTENCY**
Migrated two legacy earthquake reporting systems to **Docker + LAMP / MERN stacks**, improving deployment consistency, rollback capability and maintainability across HKO infrastructure.



Reliability Improvement 📉 **REDUCING INCIDENT RECURRENCE**
Reduced recurring system incidents through proactive monitoring, root cause analysis (RCA) and preventive fixes, directly enhancing HKO platform stability and user trust.



Documentation Standardisation 📖 **TEAM-WIDE ADOPTION**
Authored **comprehensive user guides and technical runbooks** for 3+ internal applications, adopted as the official team standard.



Cybersecurity Upskilling 🛡️ **SECURITY-FIRST MINDSET**
Completed **CompTIA Network+** and **ISC2 CISSP** core curriculum.